Organizing Fiscal Capacity*

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Abstract

This paper investigates how the organization of a tax authority's presence across the national territory impacts fiscal capacity. I examine a reform of the Brazilian tax authority, which led to a more centralized structure with fewer, larger local offices. Tax revenues decreased by 3.4% in municipalities previously served by a closed office. At the same time, tax revenues increased by 4.4% in municipalities already served by an office which became larger after the reform. A more centralized structure improves the tax authority effectiveness, but it makes it harder to monitor the local economy. The negative impact of closing offices on tax revenues is stronger in municipalities where local information is more valuable. Increased distance from the local office makes it harder to gather local information.

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1 — Introduction

Governments in low- and middle-income countries struggle to increase their tax revenues (Besley and Persson 2013, 2014). One way to overcome this challenge is to enhance the effectiveness of tax authorities, the agencies in charge of tax administration. Recent work has emphasized how their effectiveness is impacted by the quality of tax agents and by the incentives they face (Bergeron et al. 2022; Khan, Khwaja, and Olken 2016). However, even though these agencies are often large and complex bureaucracies, little is known about how the organization of their presence across the national territory impacts fiscal capacity.

An important decision facing tax authorities is the choice between a decentralized structure, with many small local offices, and a more centralized structure, with few large local offices. A decentralized structure may enable tax agents to monitor taxpayers more effectively (Scott 1998; Lee and Zhang 2017). The key trade-off is that a centralized structure can increase communication costs between tax agents and exacerbate agency problems. Having larger office may make it easier for tax agents to obtain help from more knowledgeable colleagues when handling complex cases. Similarly, it may allow the tax authority to more easily monitor agents to ensure they exert effort and do not misbehave.

In order to shed light on this trade-off, this paper studies a large reorganization of the Brazilian tax authority. I leverage variation in how different areas are affected by this reform to provide evidence on both the costs and benefits of a more centralized structure.

The Brazilian tax authority (*Receita Federal do Brasil*) is an ideal context to study the trade-off between a centralized and decentralized structure. It is responsible for the collection of around 55% of the government tax revenues, including income tax, social security contributions, and taxes on firms' inputs and profits. It employs more than 20,000 agents and is tasked with tax administration over a territory of more than 8,510,000 km^2. The entire territory is partitioned in 94 tax jurisdictions, each managed by a local office. Among other things, these local offices are tasked with detecting evasion. They do so by combining desk audits based on available data and physical inspections to collect additional information. A more decentralized structure may enable tax agents to collect additional information more easily, for instance because it reduces the distance they have to travel to visit taxpayer's premises. On the other hand, a more centralized

structure may reduce communication costs and make it easier for tax agents to access help from more knowledgeable colleagues on how to handle complex cases. Furthermore, a centralized structure can allow the organization to better monitor the agents, and ensure they exert effort and do not misbehave.

The authority underwent a significant reorganization in 2020. The reform was triggered by mandated budget cuts, which led to a hiring freeze, a decline in personnel size, and a reduction in the number of maintainable managerial positions. The reform has two components: it closes some local offices and expands others. One-fourth of the existing local offices were closed. One-fourth of the existing local offices were instead expanded. Municipalities served by the closed offices were reassigned to the jurisdiction of these expanded offices. These offices also absorbed most of the tax agents from the closed offices. The reform resulted in a more centralized structure, with fewer, larger local offices.

To empirically investigate the impact of this reform, I combine various data sources to track the evolution of the tax authority presence and effectiveness across the national territory over the period 2016 to 2022. I reconstruct changes of tax offices' locations and tax jurisdictions boundaries from internal regulations. I measure effectiveness using data on the amount of federal tax revenues collected in each of the 5,565 municipalities - the lowest administrative level in Brazil. Moreover, I leverage novel administrative data tracing tax agents across offices and their trips for work to investigate the mechanisms. The resulting dataset allows me to measure the effects of the reform at a geographically disaggregated level.

Municipalities can be grouped into three categories based on how they are affected by the reform. The first group includes municipalities previously served by a local office that was closed — the *closing* municipalities. The second group consists of municipalities already served by an office that absorbed staff and jurisdiction from a closed office — the *receiving* municipalities. The third group comprises municipalities whose local offices neither closed nor became larger — the *unaffected* municipalities. If having fewer offices deteriorates tax enforcement, tax revenues in *closing* municipalities should be negatively affected. If having larger offices improves the tax authority effectiveness, tax revenues in both *closing* and *receiving* municipalities should be positively affected.

In the first part of the paper, I provide evidence on how the reform impacted tax revenues. My

empirical strategy compares the evolution of tax revenues between municipalities in different groups to provide evidence on both costs and benefits of a centralized structure. Because economic structure in Brazil is extremely heterogeneous and the choice of which office to close is not done at random, these three groups of municipalities are on differential trends in economic activity and tax revenues before the reform. To overcome this issue, I employ a matched difference-in-differences design. I match each unit in the treated group - either the *closing* or the *receiving* municipalities - to a unit from the *unaffected* municipalities based on the evolution of local GDP in the years before the reform. This allows me to compare the evolution of tax revenues among groups with similar economic activity trajectory before the reform. The assumption is that groups on similar trends in economic activity are also on similar trends in tax revenues.

To investigate the costs associated with a more centralized structure, my difference-in-differences design compares the evolution of tax revenues between municipalities whose jurisdiction tax office closed - *closing* municipalities - and *unaffected* municipalities which were on similar economic trends before the reform. I find that *closing* municipalities experience a 3.4% reduction in federal tax collection. I then turn to study whether a more centralized structure leads to some benefits. I compare the evolution of tax revenues between municipalities whose jurisdiction tax office absorbed the jurisdiction and tax agents from a close office - *receiving* municipalities - and *unaffected* municipalities which were on similar economic trends before the reform. I find that *closing* municipalities experience a 4.4% increase in federal tax collection. In both comparisons, the absence of differential trends in tax revenues before the reform lends support to the credibility of the parallel trends assumption and to the validity of the empirical design.

Having shown that the reform impacts tax enforcement differentially in *closing* and *receiving* municipalities, the second part of the paper investigates what drives these effects. While tax agents' ability to detect evasion using technology and hard data should be unaffected by their location, their ability to gather local information may be. Therefore, we might expect stronger effects of office closings on enforcement in areas where local information is more valuable. Local information is particularly relevant in settings with less third-party reporting (Jensen 2022). In Brazil, inspections by the Ministry of Labor ensure that firms register their employees, creating an information trail on employees' income and firms' activities. This helps tax agents detect evasion even without access to local information. I use microdata on labor inspections from the

years before the reform to classify municipalities based on the intensity of third-party reporting during that period. Heterogeneity analysis shows that office closings have a stronger effect in municipalities where third-party reporting is less prevalent and local information is thus more valuable

Why do the closings make it harder to collect local information? One possibility is that tax officials are now farther from the municipalities they need to monitor, and this distance makes it more difficult to gather local information. Indeed, I document that the reform increases the travel time between a *closing* municipality and the jurisdiction's tax office by an average of 1.5 hours. I then use an instrumental variable approach to isolate the effect of distance on tax revenues.

My empirical strategy leverages the baseline variation in distance between a municipality and the closest *other* office within the same fiscal region. This distance does not affect tax enforcement before the reform, as what matters is the distance to the actual jurisdiction's tax office. However, for *closing* municipalities, this distance becomes the lower bound on the distance to the actual jurisdiction's tax office after the reform. Using this approach, I find that municipalities that end up one hour farther from a tax office experience a 1.3% reduction in tax revenues.

One reason why distance may matter is that it makes it costlier for tax agents to visit municipalities farther away. While I do not have data on physical inspections, I proxy them using data on the trips for work carried out by tax agents. I find that municipalities farther away receive fewer visits from tax agents. This suggests that distance acts as a friction preventing tax agents from gathering local information.

Taken together, these findings provide some of the first evidence on how the organization of a bureaucracy presence across the national territory impacts its effectiveness. The paper shows that a more centralized structure entails costs for tax enforcement: having fewer offices makes it harder to collect local information. At the same time, a centralized structure improves the effectiveness of the tax authority: being served by larger offices leads to an increase in tax revenues. More generally, by uncovering the forces underlying the trade-off between a centralized or decentralized structure, this paper suggests that policymakers interested in implementing a similar reform should consider how relevant acquiring local information is in their context and whether geographical frictions can make that process more difficult.

Related Literature This paper relates and contributes to various strands of the literature. First, it speaks to the literature on tax administration in developing countries (Pomeranz and Vila-Belda 2019; Okunogbe and Tourek 2024). Past work has studied the role of incentives for tax collectors (Khan, Khwaja, and Olken 2016, 2019), the type of agent hired (Balán et al. 2022), specialization (Basri et al. 2021), and the potentialities of technology (Okunogbe and Santoro 2023; Dzansi et al. 2022; Knebelmann, Pouliquen, and Sarr 2023). This paper shows that the decision on how to organize the presence of tax authorities across a territory can affect their effectiveness.

Second, this paper is related to the literatures on the importance of third-party reporting (Pomeranz 2015; Naritomi 2019; Jensen 2022) and on the value of local information (Balán et al. 2022; Knebelmann, Pouliquen, and Sarr 2023) for taxation. It shows how a tax authority's ability to collect local information is constrained by the distance from taxpayers, and highlights that this ability is especially important in areas with limited third-party reporting.

Third, this paper contributes to the literature public sector personnel (Finan, Olken, and Pande 2017). While previous literature has investigated the role of selection (Dal Bó, Finan, and Rossi 2013; Ashraf et al. 2020), incentives (Bertrand et al. 2020), and managers (Fenizia 2022), this paper follows Besley et al. 2022 call to use microdata for examining the effects of system-wide reforms and the importance of different organizational designs. Since the reform examined here alters the size of tax jurisdictions, this paper also connects to the literature on decentralization which explores the effects of different polity sizes (Bardhan 2002; Narasimhan and Weaver 2023; Dahis and Szerman 2024).

Finally, this paper speaks to the literature on the relationship between distance from capitals and state capacity (Mann 1984; Herbst 2014; Müller-Crepon 2023). By focusing on a specific dimension of state capacity and leveraging detailed information on the locations of bureaucrats performing this function, I am able to provide causal evidence on how state capacity deteriorates with distance.

Outline The rest of the paper is organized as follows: Section **??** describes the context and the tax authority reform, while section **??** introduces the data. Section **??** presents evidence on the effects of the reform on tax revenues, whereas Section **??** investigate the mechanisms behinds

these effects. In Section **??** I perform a back of the envelope calculation of the net effects of the before, before concluding in Section **??**.

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